

Lower Universal Anchorage Systems for Restraint Systems and Booster Seats (Standard 210.2)

- [SOR/2013-117, s. 9]

Application

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210.2 (1) Subject to subsection (2), this section applies to every

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- (a) passenger car;

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- (a.1) three-wheeled vehicle;

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- (b) multi-purpose passenger vehicle and truck with a GVWR of 3 856 kg or less and an unloaded vehicle weight of 2 495 kg or less;

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- (c) bus, other than a school bus, with a GVWR of 4 536 kg or less; and

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- (d) school bus.

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(2) This section does not apply to

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- (a) a walk-in van;

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- (b) a bus that has only one row of forward-facing designated seating positions rearward of the driver's seat;

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- (c) a vehicle that is not equipped with a manual cut-off switch to deactivate the frontal air bag installed at the right front outboard designated seating position when any restraint system or booster seat is installed, and that

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- (i) has only one row of forward-facing designated seating positions,

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- (ii) cannot accommodate, because of the location of transmission or suspension components, the installation of the bars of a lower universal anchorage system to permit the attachment to them of a child restraint fixture as shown in Figure 1 in a seating position located to the rear of the first row of seating positions, or

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- (iii) has no seating position located to the rear of the first row of seating positions that is able to accommodate a child restraint fixture shown in Figures 2 to 4; or

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General

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(3) A lower universal anchorage system consists of two straight horizontal bars that

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- (a) are transverse to the longitudinal axis of the vehicle;

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- (b) have a diameter of $6 \text{ mm} \pm 0.1 \text{ mm}$ and a length, measured in accordance with Figure 12, of not less than 25 mm;

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- (c) are parallel, with collinear centroidal longitudinal axes, and are spaced laterally so that they permit the lower connectors on a child restraint fixture, shown in Figures 2 to 4, to be attached to them over the entire length of the lower connectors;

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- (d) are an integral and permanent part of a vehicle seat or seat anchorage or the structure of the vehicle such that they can be removed only with the use of tools;

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- (e) are rigidly attached to a vehicle seat or seat anchorage or the structure of the vehicle so that they will not deform more than 5 mm when subjected to a force of 100 N in any direction with the vehicle seat adjusted in the vehicle to its full rearward and full downward position and the seat back adjusted in the nominal design riding position; and

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(f) permit a checking device to be attached to them over the entire width of the checking device, with a gap of less than 1 mm between the surface of the bars and line M, shown in Figure 5.

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(4) Subject to subsections (7) and (8), a lower universal anchorage system shall be installed in a vehicle

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(a) in the case of a vehicle, other than a school bus, that has only one row of forward-facing designated seating positions, at one forward-facing designated seating position other than that of the driver;

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(b) in the case of a vehicle, other than a school bus, that has two or more rows of designated seating positions, and not more than two forward-facing designated seating positions located to the rear of the first row of designated seating positions, at all forward-facing designated seating positions located to the rear of the first row of designated seating positions;

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(c) in the case of a vehicle, other than a school bus, that has two or more rows of designated seating positions, and three or more forward-facing designated seating positions located to the rear of the first row of designated seating positions, at two forward-facing designated seating positions located to the rear of the first row of designated seating positions, with at least one lower universal anchorage system being installed at a forward-facing designated seating position in the second row of designated seating positions;

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(d) in the case of a school bus that has not more than 24 passenger designated seating positions, and only one forward-facing designated seating position other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at that forward-facing designated seating position;

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(e) in the case of a school bus that has not more than 24 passenger designated seating positions, and two or more forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at two of those forward-facing designated seating positions;

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(f) in the case of a school bus that has 25 or more, but not more than 65, passenger designated seating positions, and not more than three forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at all of those forward-facing designated seating positions;

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(g) in the case of a school bus that has 25 or more, but not more than 65, passenger designated seating positions, and four or more forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at four of those forward-facing designated seating positions;

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(h) in the case of a school bus that has 66 or more passenger designated seating positions, and not more than seven forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at all of those forward-facing designated seating positions; and

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(i) in the case of a school bus that has 66 or more passenger designated seating positions, and eight or more forward-facing designated seating positions other than that of the driver and those of a bench seat that is next to an emergency exit and contains an adjacent seat, at eight of those forward-facing designated seating positions.

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(5) [Repealed, SOR/2008-72, s. 9]

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(6) A lower universal anchorage system shall be installed only at a designated seating position that is equipped with a user-ready tether anchorage, except in the case of a convertible or an open-body type vehicle.

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(7) The number of lower universal anchorage systems required in a vehicle under subsection (4) may be reduced by the number of built-in restraint systems installed in the vehicle.

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(8) Except in the case of a bus, if the distance between the rearward surface of the front seat back of the vehicle and the forward surface of the rear seat back of the vehicle is less than 720 mm, as measured in accordance with Figure 6, and if the vehicle is equipped with the manual cut-off switch referred to in paragraph (2)(c), a lower universal anchorage system may be installed at a passenger designated seating position in the first row of designated seating positions instead of at a designated seating position located to the rear of the first row of designated seating positions.

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(9) A vehicle that is equipped with a seat that slides sideways or that can be installed in any other seating position in the vehicle shall meet the requirements of this section with the seat adjusted in any adjustment position.

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(10) [Repealed, SOR/2008-72, s. 9]

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Lower Universal Anchorage System Positioning

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(11) Each bar of the lower universal anchorage system shall be installed so that a child restraint fixture can be attached to the system as shown in Figure 1.

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(12) For the purposes of subsection (11), the vehicle seat shall be adjusted to its full rearward and full downward position, with the seat back in the nominal design riding position and the child restraint fixture shown in Figures 2 and 3 or in Figure 4 installed in the seat.

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Strength Requirements

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(13) A lower universal anchorage system installed in a row of designated seating positions shall not separate completely from the vehicle seat or seat anchorage or the structure of the vehicle when tested

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(a) subject to subsection (15), by pulling with a force of 15 000 N, which force is

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(i) applied by means of a test device shown in Figures 7 and 8 that is installed using both the associated user-ready tether anchorage and the lower universal anchorage system as a child restraint system would be installed in accordance with the vehicle manufacturer's instructions,

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(ii) applied in a forward direction parallel to the vehicle's vertical longitudinal plane through the X point on the test device, starting with a pre-load force of 500 N,

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(iii) applied initially, along a line above the horizontal line, at an angle of $10^\circ \pm 5^\circ$ to it,

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(iv) attained within 30 seconds, at any onset force rate of not more than 135 000 N/s, and

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(v) maintained at a level of 15 000 N for a minimum of one second; and

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(b) by pulling with a force of 5 000 N, which force is

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(i) applied by means of a test device shown in Figures 7 and 8 that is installed as a child restraint system would be installed in accordance with the vehicle manufacturer's instructions using both the user ready tether anchorage and the lower universal anchorage system or, at the manufacturer's option, using only the lower universal anchorage system,

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(ii) applied along a vertical longitudinal plane that is at an angle of $75^\circ \pm 5^\circ$ to either side of a vertical longitudinal plane that is parallel to the vehicle's longitudinal centre line, through the X point on the test device, starting with a pre-load force of 500 N,

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(iii) applied initially, along a horizontal line or along any line below or above that line that is at an angle to that line of not more than 5° ,

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(iv) attained within 30 seconds, at any onset force rate of not more than 135 000 N/s, and

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(v) maintained at a level of 5 000 N for a minimum of 1 second.

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(14) When a force of 5 000 N is applied in accordance with paragraph (13)(b), the lower universal anchorage system shall not permit the X point on the test device to be displaced in the direction of the applied force by more than 125 mm if the test device is installed in an outboard designated seating position or by more than 150 mm if the test device is installed in an inboard designated seating position.

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(15) A lower universal anchorage system installed in a designated seating position that is not equipped with an associated user-ready tether anchorage shall be tested by the application of a force of 11 000 N that is

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(a) applied by means of a test device shown in Figures 7 and 8 that is installed using only the lower universal anchorage system;

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(b) applied in a forward direction parallel to the vehicle's vertical longitudinal plane through the X point on the test device, starting with a pre-load force of 500 N;

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(c) applied initially, along a line above the horizontal line, at an angle of $10^\circ \pm 5^\circ$ to it;

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(d) attained within 30 seconds, at any onset force rate of not more than 135 000 N/s; and

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(e) maintained at a level of 11 000 N for a minimum of one second.

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Test Requirements

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(16) If a bench seat in a bus or a row of designated seating positions in another vehicle has more than one lower universal anchorage system and a distance of 400 mm or more, measured in accordance with Figure 9, separates the midpoints of any adjacent designated seating positions, the force referred to in subsection (13) or (15) shall be applied simultaneously to each anchorage system in the manner specified in that subsection.

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(17) The strength requirement tests shall be carried out under the following conditions:

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(a) the vehicle seat is installed in the vehicle;

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(b) the vehicle seat is adjusted to its full rearward and full downward position;

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(c) the vehicle seat back is adjusted in the nominal design riding position;

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(d) the head restraints, if installed at a designated seating position that is equipped with a lower universal anchorage system, are adjusted in accordance with the manufacturer's instructions; and

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(e) a rearward force of $135\text{ N} \pm 15\text{ N}$ is applied to the centre of the lower front crossbar of the test device shown in Figures 7 and 8 to press the test device against the seat back as the position of the lower connectors of the test device is adjusted to remove slack or tension.

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(17.1) Despite subsection (17), in the case of a bus, the strength requirement tests may be carried out when the vehicle seat is not installed in the vehicle if

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(a) the user-ready tether anchorage is installed on the vehicle seat in accordance with the tether anchorage positioning requirements of section 210.1 for a designated seating position of the vehicle;

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(b) the structure on which the seat is installed, the seat anchorage and all associated hardware are equivalent to those used when the vehicle seat is installed in the vehicle; and

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(c) a rearward force of $135\text{ N} \pm 15\text{ N}$ is applied to the centre of the lower front crossbar of the test device shown in Figures 7 and 8 to press the test device against the seat back as the position of the lower connectors of the test device is adjusted to remove slack or tension.

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Information Respecting Lower Universal Anchorage Systems

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(18) Subject to subsections (19) and (20), if a lower universal anchorage system is not visible along a line making an upward angle of at least 30° with the horizontal plane without compression of the seat cushion or seat back of the seating position, the presence of each bar of the system shall be indicated by the symbol shown in Figure 10, consisting of a circle containing a pictogram, which symbol shall meet the following conditions:

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- (a) the pictogram shall contrast with the background of the circle;

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- (b) the circle shall contrast with its background, namely, the seat back or seat cushion of the vehicle;

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- (c) its centre shall be located no more than 25 mm from the vertical longitudinal plane that passes through the centre of each bar of the system;

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- (d) it shall be marked on the seat back or on the seat cushion of the vehicle in accordance with Figure 11; and

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- (e) it shall be permanently marked by any means in a manner that makes it impossible to remove without defacing or destruction.

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(19) In the case of a foldable lower universal anchorage system, its presence shall be indicated by the symbol shown in Figure 10 marked in accordance with subsection (18) or marked on the mechanism for deploying each bar of the foldable system and visible at the junction of the seat back and the seat cushion when the system is in the folded position.

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(20) In the case of a lower universal anchorage system hidden by one or more covers, its presence may be indicated by the symbol shown in Figure 10 marked on each cover in the following cases:

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(a) the covers are not entirely removable; or

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(b) the system is visible in accordance with subsection (18) if the covers are entirely removed.

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(21) [Repealed, SOR/2008-72, s. 9]

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(22) The English and French versions of the owner's manual for a vehicle that is equipped with a lower universal anchorage system shall contain the following information:

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(a) if the symbol shown in Figure 10 is marked on a seat back, seat cushion or cover over the lower universal anchorage system, an explanation that the symbol indicates the presence of a lower universal anchorage system, and a reproduction of the symbol;

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(b) [Repealed, SOR/2008-72, s. 9]

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(c) an identification, by means of the symbol shown in Figure 10 or any other mark used to indicate the presence of a lower universal anchorage system, as applicable, of the designated seating positions that are equipped with a lower universal anchorage system;

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(d) if the vehicle is equipped with a foldable system, instructions on how to deploy the system;

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(e) if the vehicle is equipped with head restraints at the seating positions equipped with a lower universal anchorage system, instructions on the correct manner of adjusting the head restraints for use with user-ready tether anchorages;

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(f) if the vehicle has more than one row of designated seating positions, a statement that children and infants are safer when properly restrained in a child restraint system or infant restraint system secured in a rear seating position; and

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(g) if the vehicle is a convertible or an open-body type vehicle that has no designated seating position equipped with a user-ready tether anchorage, a statement that neither a restraint system nor a booster seat requiring the use of a tether strap can be properly secured to the vehicle.

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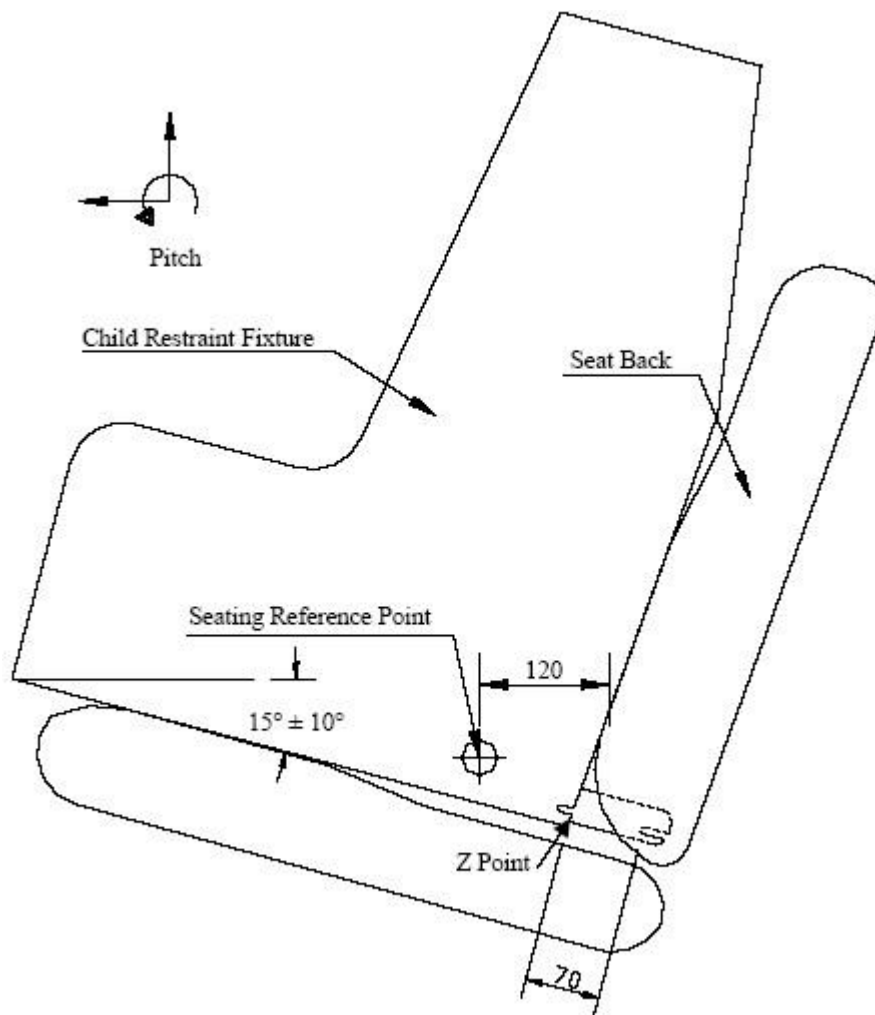


Figure 1 — Lower Universal Anchorage System Positioning

Notes

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Dimensions in mm, except where otherwise indicated

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Drawing not to scale
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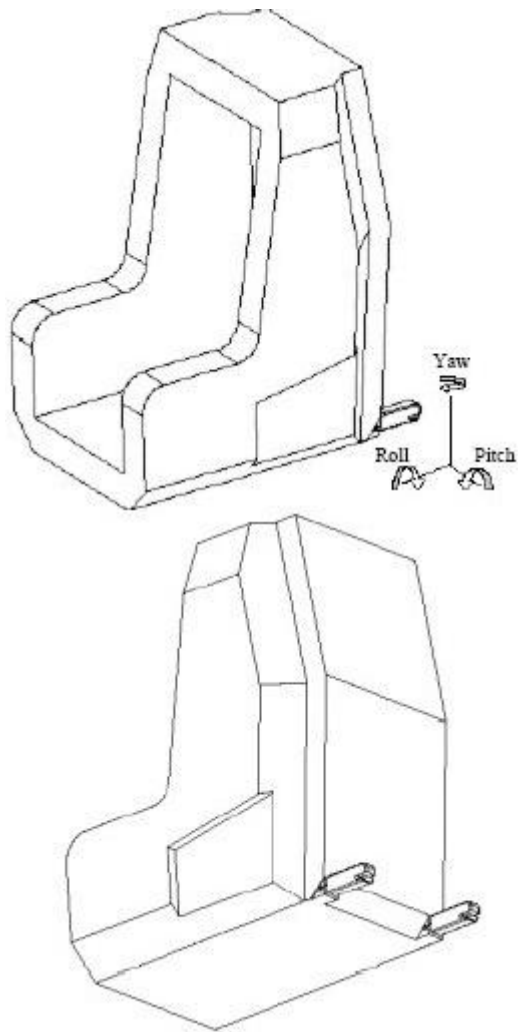


Figure 2 — Three-dimensional Schematic Views of Child Restraint Fixture

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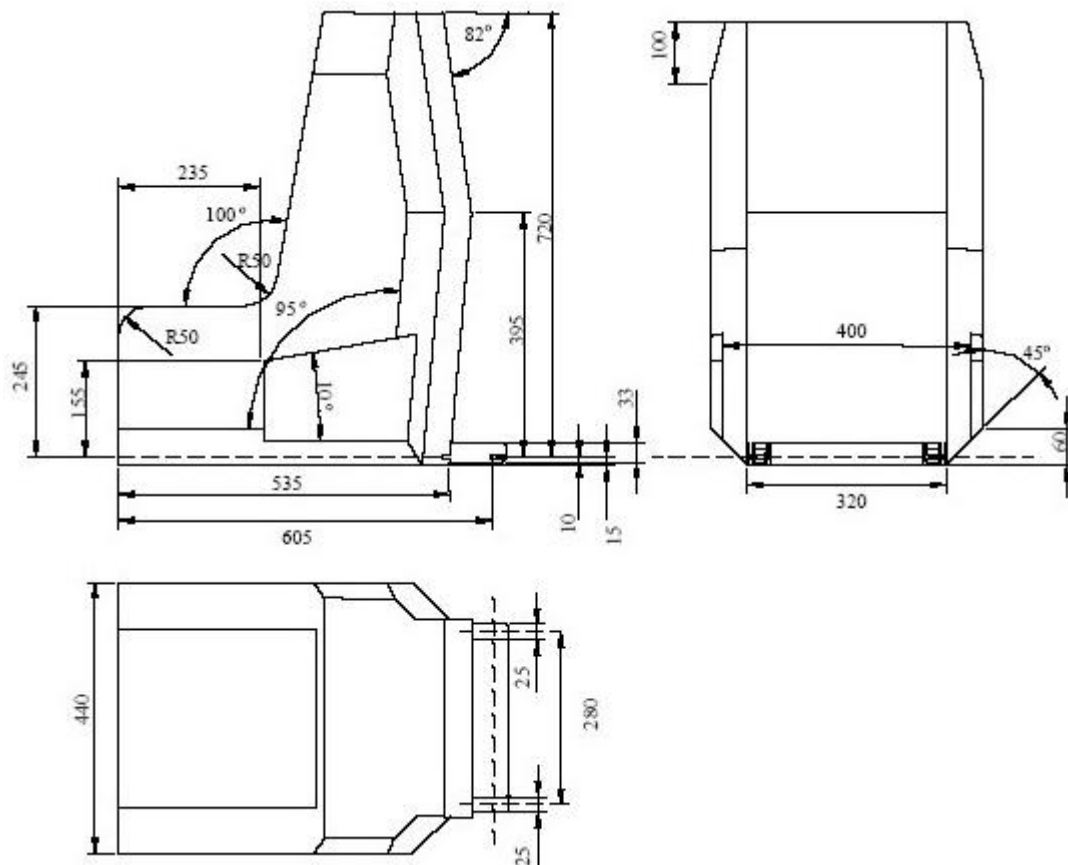


Figure 3 — Side, Back and Top Views of Child Restraint Fixture

Notes

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- 2

Dimensions in mm, except where otherwise indicated

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- 2

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- 2

Drawing not to scale

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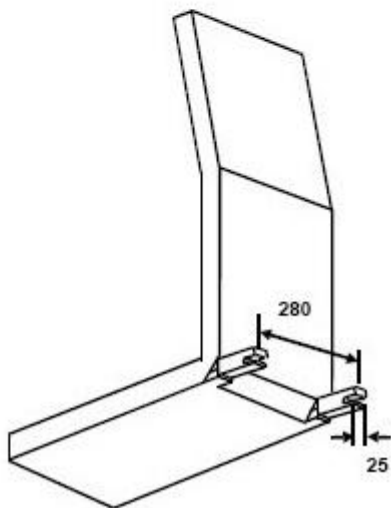
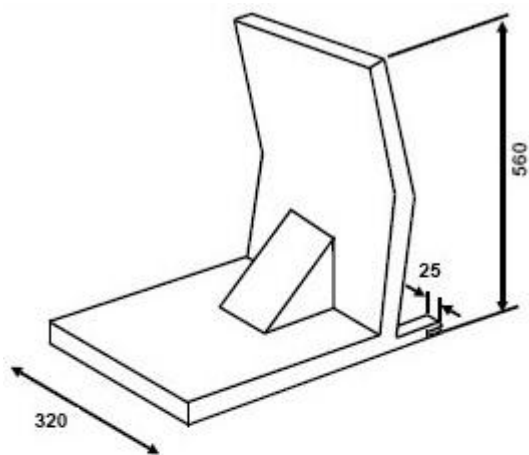


Figure 4 — Three-dimensional Schematic Views of Child Restraint Fixture with Side and Top Portals Removed

Notes

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Dimensions in mm, except where otherwise indicated

2

Drawing not to scale

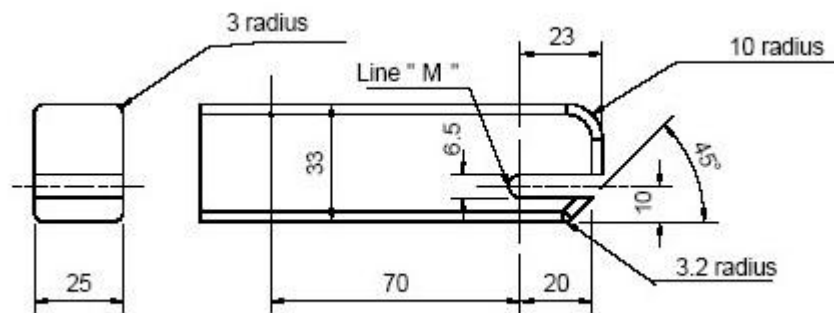


Figure 5 — Rear and Side View of Checking Device for Lower Universal Anchorage System — Envelope Dimensions

Notes:

1

Dimensions in mm, except where otherwise indicated

2

Drawing not to scale

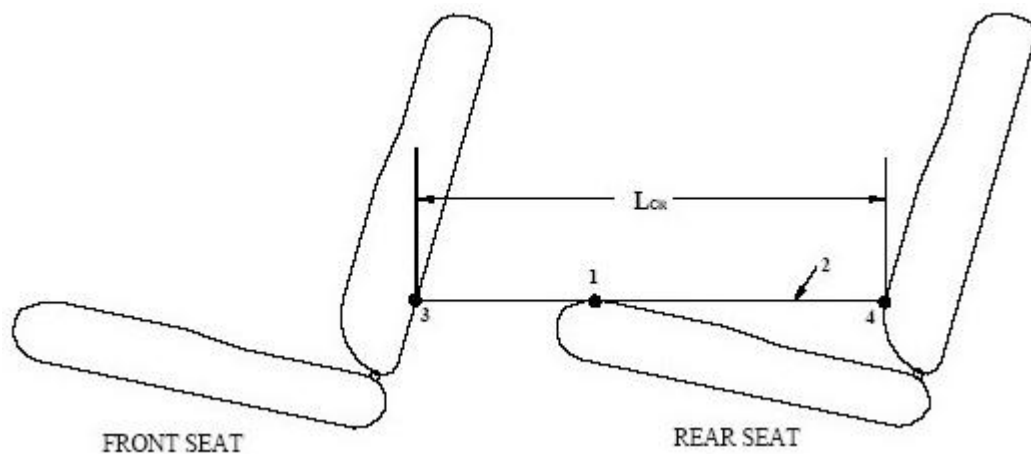


Figure 6 — Distance Between Rows of Designated Seating Positions

Legend

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1

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Highest point of the rear seat cushion in the vertical longitudinal plane that passes through the center of the driver's seat cushion, if the vehicle is equipped with front bucket seats, or through the center of the steering wheel rim, if the vehicle is equipped with a front bench seat

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2

•

Longitudinal horizontal line tangent to point 1, extending between point 3 and point 4. The length of line 2 is defined as LCR

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•
3

•

Rearward surface of the front seat back along line 2

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4

Forward surface of the rear seat back along line 2

5

LCR - The distance between the rearward surface of the front seat back and the forward surface of the rear seat back

Notes

1

Adjustable seats are in the adjustment position midway between the forwardmost and rearmost positions and, if separately adjustable in a vertical direction, are at the lowest position. If an adjustment position does not exist midway between the forwardmost and rearmost positions, the closest adjustment position to the rear of the midpoint is used.

2

Place adjustable seat backs in the nominal design riding position as recommended by the manufacturer, in the manner specified by the manufacturer. Place adjustable anchorages in the nominal design riding position recommended by the manufacturer for a 50th percentile adult male occupant. Place each adjustable head restraint in its highest adjustment position. Adjustable lumbar supports are positioned so that the lumbar support is in its least firm adjustment position.

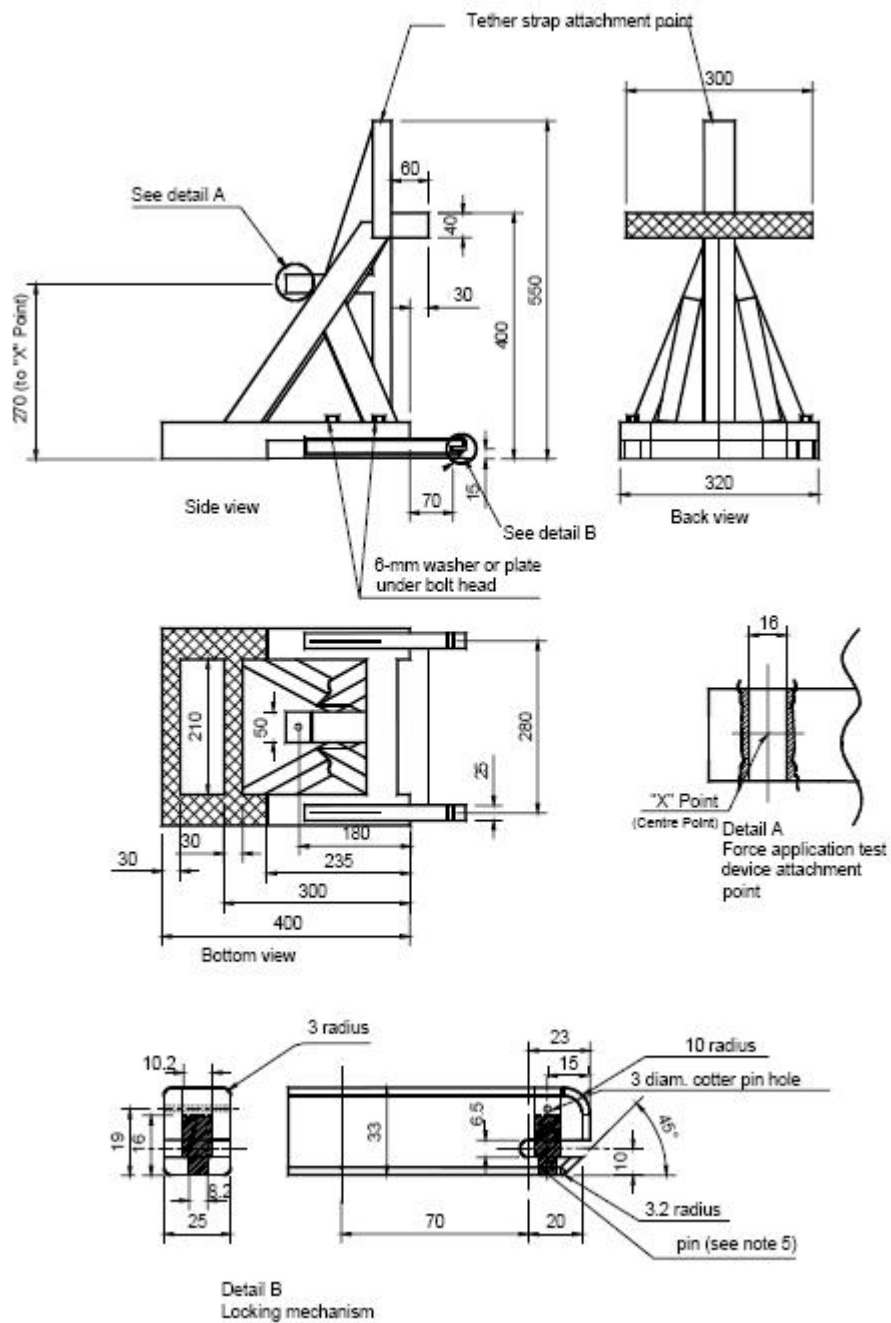


Figure 7 — Side, Back and Bottom View of the Static Force Application Test Device for Strength Requirements Test

Notes

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Dimensions in mm, except where otherwise indicated.

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2

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Drawing not to scale.

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Material: Unless otherwise indicated, steel, mild steel rectangular tubing 50 mm by 75 mm of 3 mm nominal thickness, with 6 mm thick force application test device attachment point plate.

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Securely welded construction.

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Pin dimensions: overall height: 16 mm; small diameter height: 6.75 mm; large diameter: 10 mm; small diameter: 8 mm.

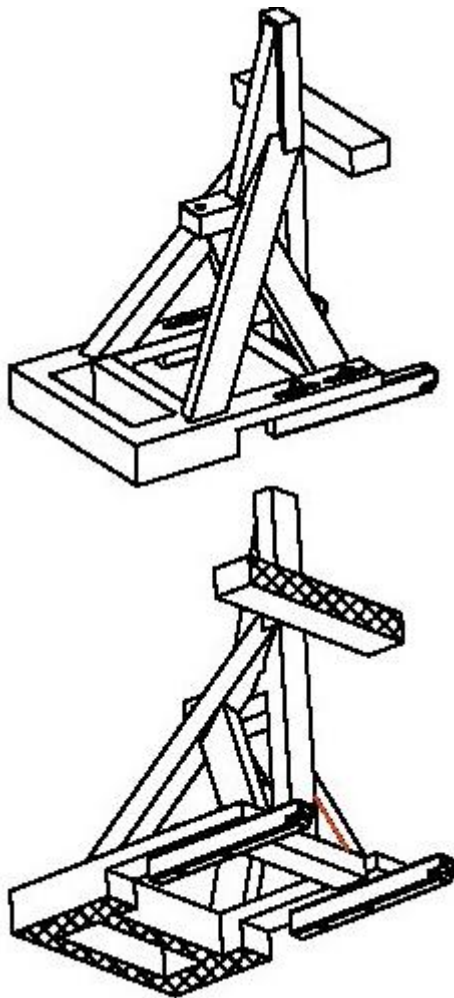


Figure 8 — Three-dimensional Schematic Views of the Static Force Application Test Device for Strength Requirements Test

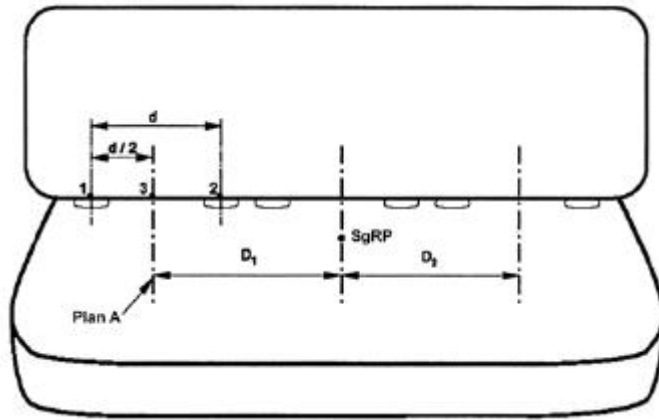


Figure 9 — Measurement of Distance Between Adjacent Designated Seating Positions for Use in Simultaneous Testing

Legend:

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 $d =$
 •
 centre to centre distance between the bars of a lower universal anchorage system for a given seating position (nominal distance of 280 mm)

•
 •
 $D =$
 •
 distance between vertical longitudinal planes located midway between the bars of a lower universal anchorage system for a given seating position

•
 •
 $SgRP =$
 •
 seating reference point

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Notes:

1

Drawing not to scale

2

The midpoint of a designated seating position lies in the vertical longitudinal plane that is equidistant from the vertical longitudinal planes through the geometric centre of each of the two bars of the lower universal anchorage system installed at the seating position. For those designated seating positions that do not have the lower universal anchorage system bars, the midpoint of a designated seating position lies in the vertical longitudinal plane that passes through the SgRP of the seating position.

3

The distance shall be measured between the vertical longitudinal planes passing through the midpoint of adjacent designated seating positions along a line perpendicular to the planes.

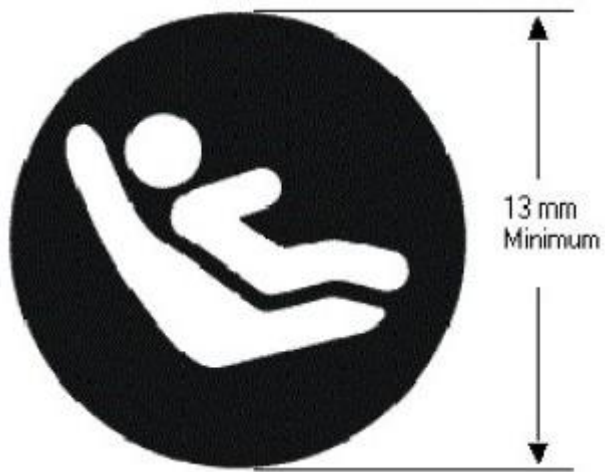


Figure 10 — Lower Universal Anchorage System Symbol

Notes

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Drawing not to scale.

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Symbol may be shown in mirror image.

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Colour of the symbol at choice of manufacturer.

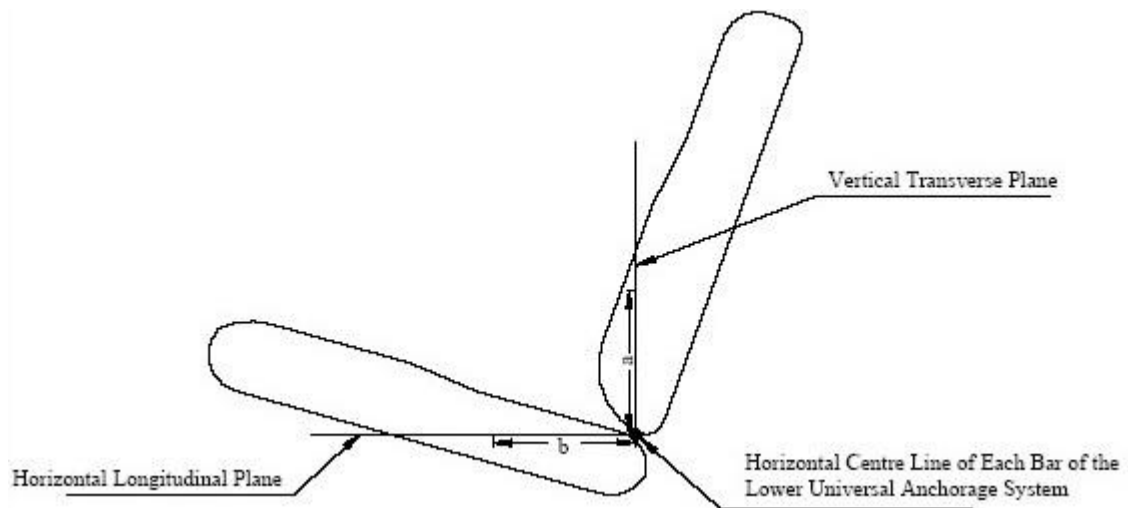


Figure 11 — Placement of Symbol on the Seat Back and Seat Cushion of a Vehicle

Notes

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- Drawing not to scale.
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- 2
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- 50 mm = a = 100 mm.
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- 3
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- b = 100 mm ± 25 mm.
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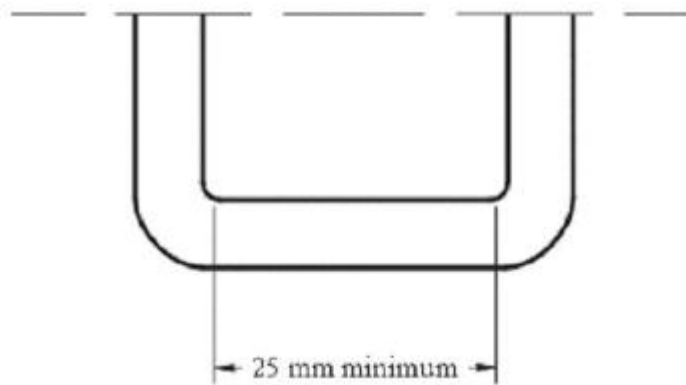


Figure 12 — Width of Lower Universal Anchorage Bar, Top View

Note: Drawing not to scale

- - SOR/2002-205, s. 4;
 - SOR/2003-272, s. 27;
 - SOR/2005-342, s. 4;
 - SOR/2006-94, ss. 3, 4(E);
 - SOR/2008-72, s. 9;
 - SOR/2008-104, s. 17;
 - SOR/2013-117, s. 10.

211 [Repealed, SOR/97-421, s. 13]

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Windshield Mounting

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212 (1) Every passenger car, other than a forward control configuration vehicle or an open-body type vehicle with a fold-down or removable windshield, shall conform to the requirements of Technical Standards Document No. 212, Windshield Mounting (TSD 212), as amended from time to time.

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(2) Every truck, bus and multi-purpose passenger vehicle with a GVWR of 4 536 kg or less, other than a walk-in van, a forward control configuration vehicle or an open-body type vehicle with a fold-down or removable windshield, shall conform to the requirements of TSD 212, as amended from time to time.

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(3) [Repealed, SOR/2014-307, s. 19]

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- SOR/80-282, s. 3;

- SOR/82-274, s. 1;

- SOR/90-387, s. 2(F);

- SOR/93-5, s. 6;

- SOR/94-692, s. 2(F);
- SOR/95-536, s. 7(F);
- SOR/97-201, s. 5;
- SOR/97-421, ss. 17, 22(F);
- SOR/2003-57, s. 2;
- SOR/2008-72, s. 10(F);
- SOR/2009-121, s. 3;
- SOR/2013-9, s. 7;
- SOR/2014-307, s. 19.